Industrial Sensor Anomaly Detection Using Time-RCD

Zero-shot detection of faults in the SKAB water-circulation rig with a pretrained Hugging Face time-series foundation model

Applied AI with Hugging Face · Final project · 30 November 2026

Problem and research question

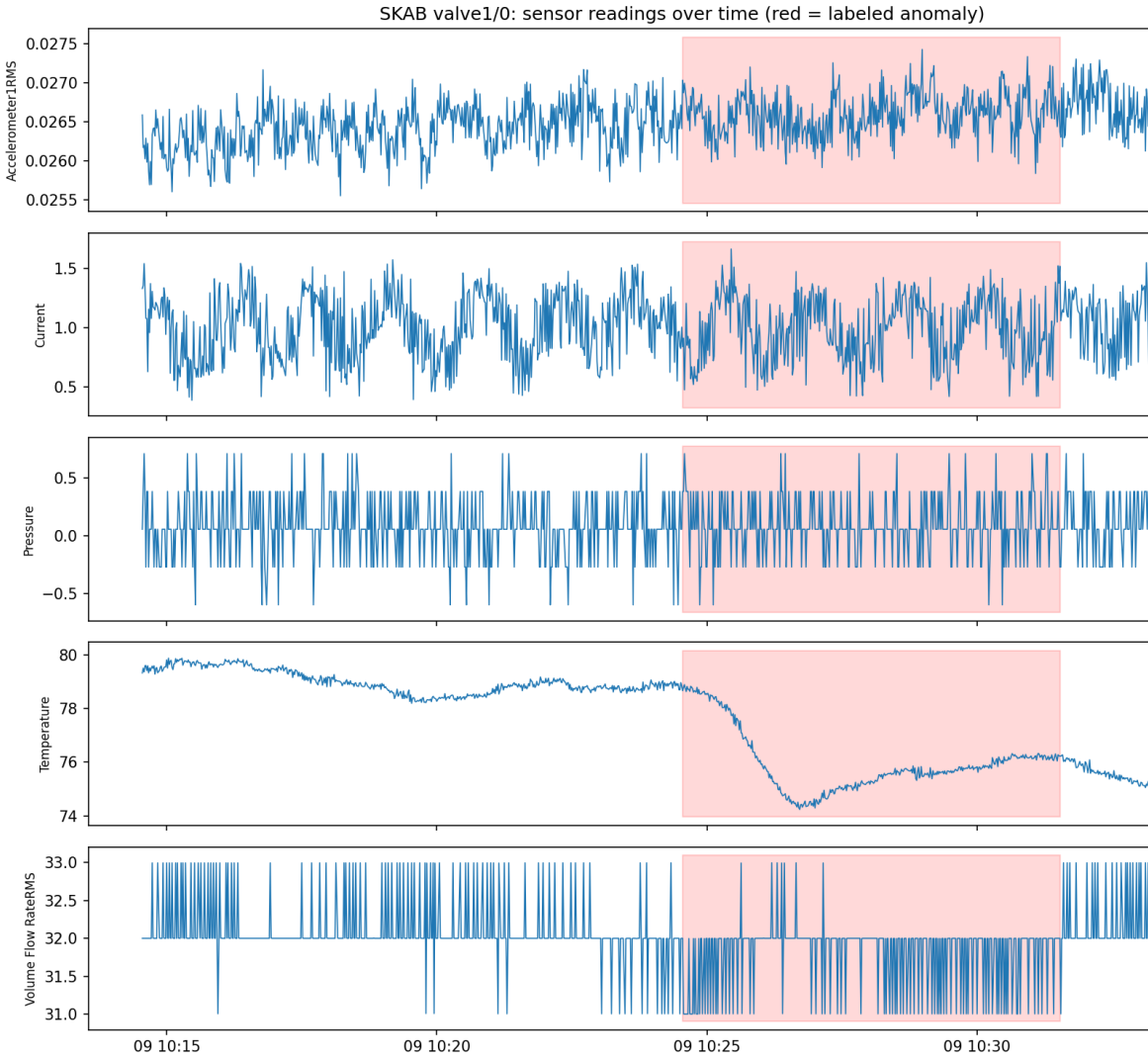
- Factory machines are monitored by many sensors; faults show up as unusual patterns over time, not as single bad values.
- Goal: an application that reads sensor CSV files, scores every second for abnormality, flags anomalous periods, and is evaluated against ground truth.
- Research question: can a pretrained zero-shot time-series foundation model detect the labeled fault periods in SKAB without any training on SKAB, and how does it compare with simple classical baselines?
- An anomaly score is a prompt for human investigation, not a diagnosis of failure.

Dataset: SKAB (Skoltech Anomaly Benchmark)

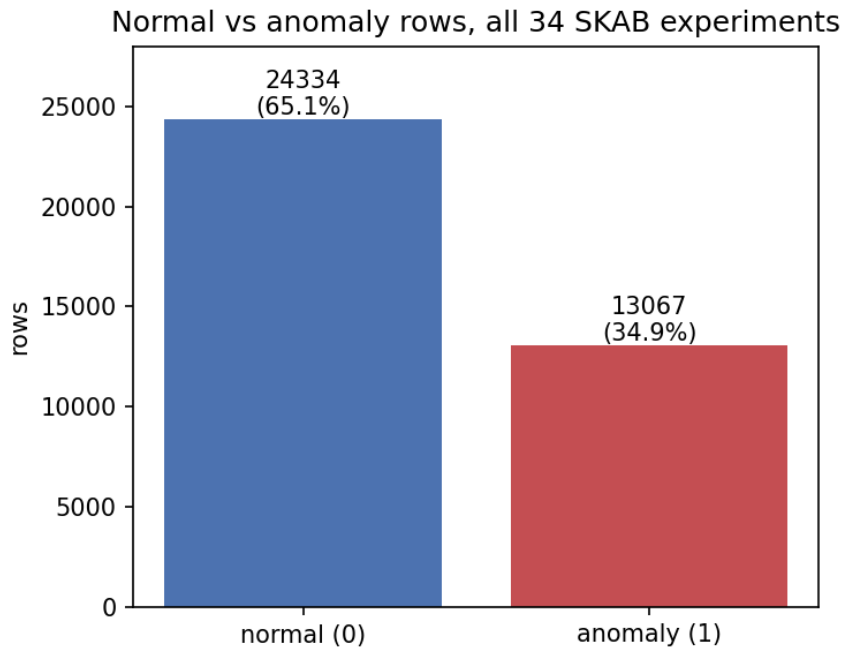
Multivariate sensor recordings from an experimental water-circulation rig · GPL-3.0

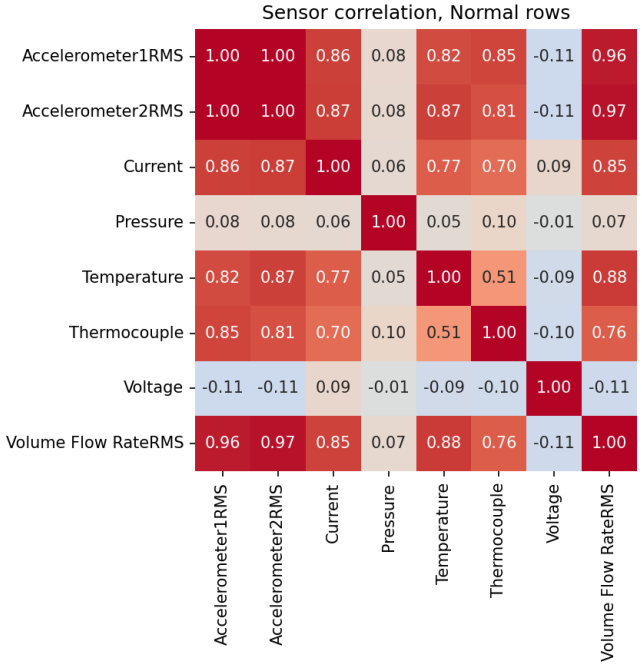
Item	Value
Labeled experiments	34 (valve1 16, valve2 4, other 14) + 1 anomaly-free
Labeled rows	37,401 at 1 Hz
Sensors	8: two accelerometers, current, pressure, two temperatures, voltage, flow rate
Labels	anomaly (0/1) and changepoint; evaluation only
Missing / duplicate rows	0 / 0

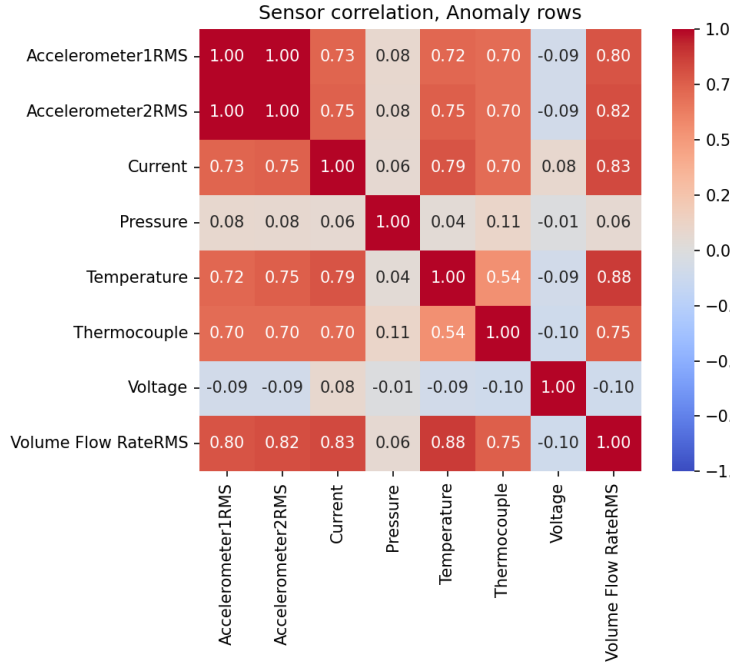
Each experiment: a few minutes of normal operation, then one induced fault (valve closing, rotor imbalance, leak, cavitation, ...) lasting 3 to 10 minutes.



EDA: three observations







- Class distribution: 34.9 % of rows are anomalous. Imbalanced but mild, because every file contains one long induced fault (25 % to 49 % per file).
- Sensor relationships: the two accelerometers are identical (r = 1.00). Flow vs accelerometer correlation drops from 0.96 in normal rows to 0.80 in anomalous rows.
- Anomaly duration: 34 segments, one per experiment, 188 to 586 s long. No point anomalies: these are collective anomalies, suited to a context-based model.

Model: Time-RCD (thu-sail-lab/Time-RCD)

Zero-shot foundation model for time-series anomaly detection · Lan et al. 2025, ICML 2026 · Apache-2.0

- Input: a series of shape (T, 8 sensors). Output: one anomaly score in [0, 1] per second.
- Idea: split the series into patches and measure how much each patch disagrees with its surrounding context (relative context discrepancy). Pretrained on synthetic data only.
- No training or fine-tuning on SKAB. One call per experiment, about 6 s on CPU.
- Multivariate checkpoint used (445 MB). The univariate checkpoint was also tested as a side experiment.
- Why this model: official Hugging Face repo, documented Python API, handles multivariate input, runs without a GPU.

```
from time_rcd import TimeRCDDetector

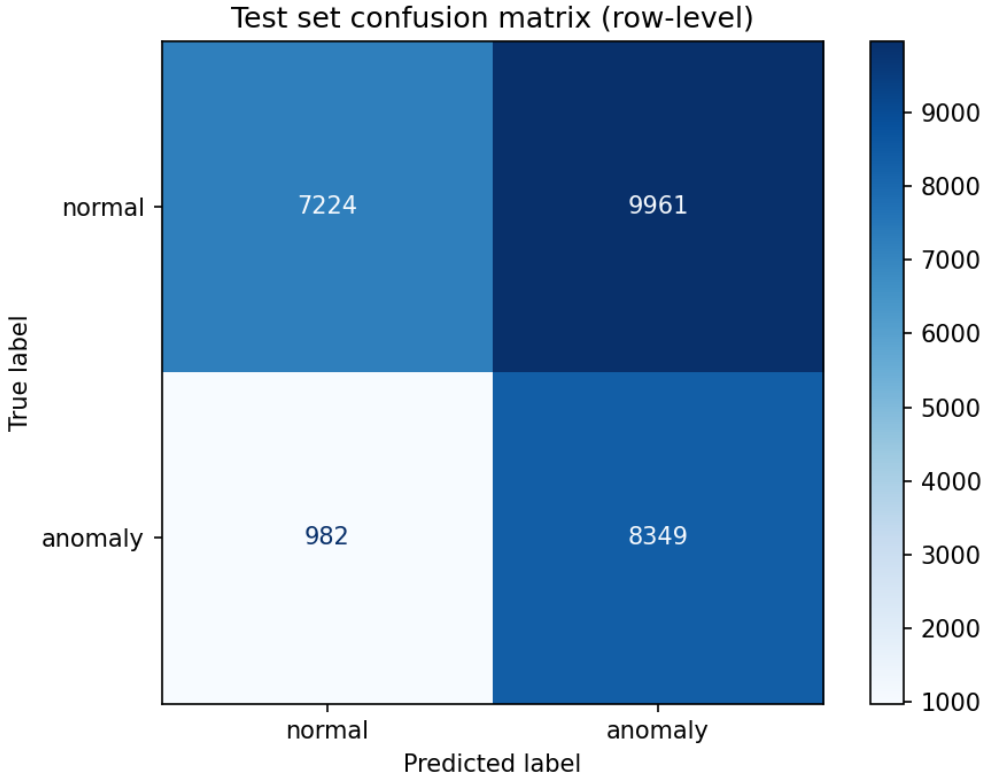
det = TimeRCDDetector.from_pretrained(
        variant="multi", device="cpu")
scores = det.predict(sensors)
# (T, 8) -> (T,)
```

Method: split, scoring, threshold, evaluation

- Split by whole experiment, fixed before any model run: 10 validation files (valve1 0-4, valve2 0, other 1/5/10/13) used only to pick the threshold; 24 test files held out. Nothing is trained, so no training split.
- Score each file with Time-RCD on the 8 raw sensor columns. Label columns are never passed to the model.
- Rank normalisation (label-free): scores are relative to each recording's own context, so each file's scores are converted to within-file percentile ranks. Pooled test ROC-AUC 0.68 to 0.74.
- Threshold: the rank that maximises F1 on validation rows (0.310), frozen, then applied to the test set.
- Metrics on test rows: accuracy, precision, recall, F1, ROC-AUC, confusion matrix, plus per-experiment metrics.
- Baselines under identical conditions: Isolation Forest fit on the anomaly-free file, rolling z-score, smoothed scores, univariate checkpoint.

Results on the 24 held-out test experiments (26,516 rows)

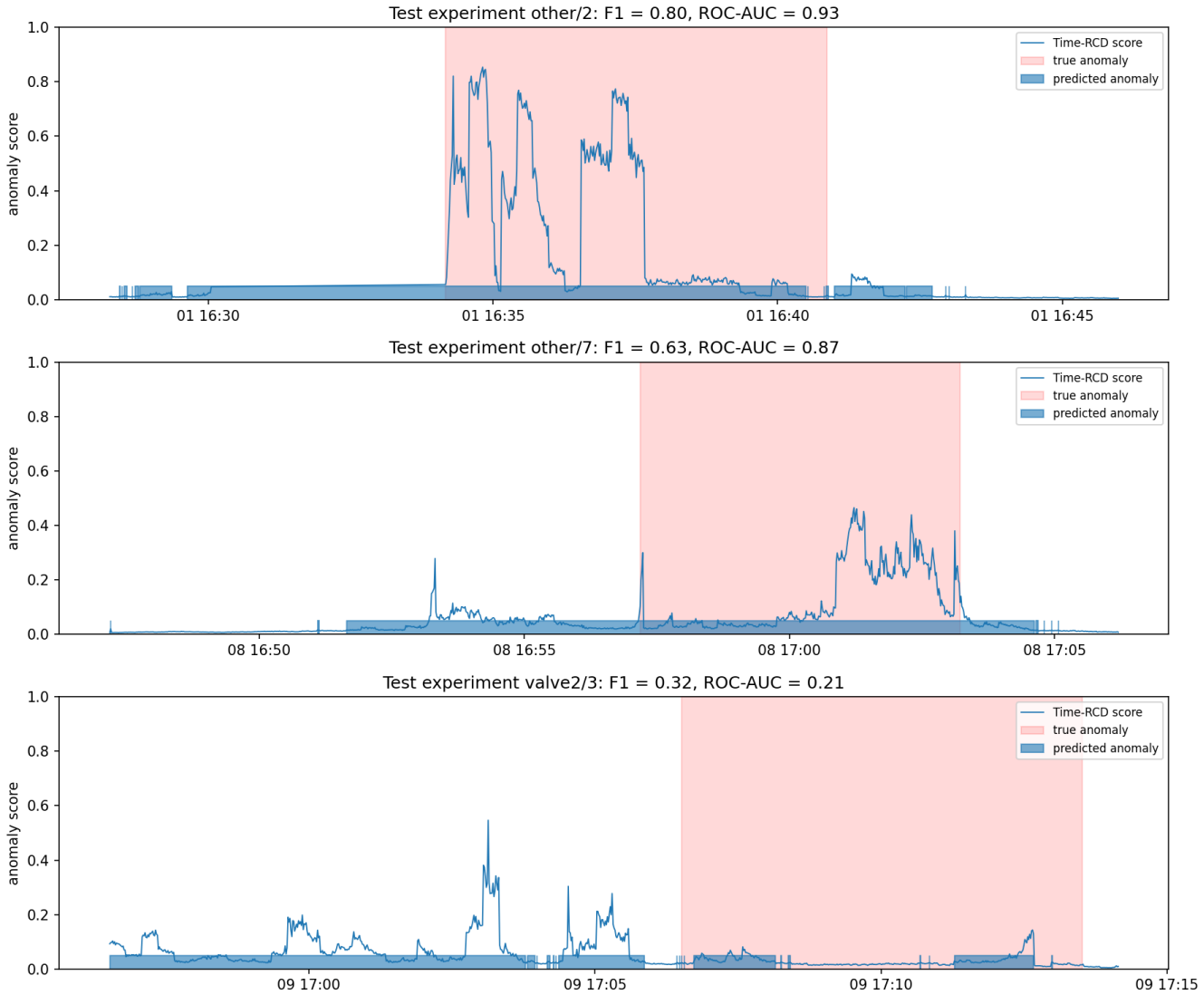
Metric	Value
Accuracy	0.587
Precision	0.456
Recall	0.895
F1	0.604
ROC-AUC (pooled)	0.737
Per-experiment ROC-AUC	median 0.79, range 0.21 to 1.00
Trivial 'all anomaly' F1	0.521 (reference)



The F1-optimal threshold favours recall: 90 % of anomalous seconds are flagged, but only 46 % of flagged seconds are truly anomalous. The model tutorial's method (raw score above the 95 % quantile of the anomaly-free file) gives the opposite trade-off: precision 0.57, recall 0.25, F1 0.34.

Model scores vs ground truth on three test experiments

A strong (other/2), a median (other/7) and a weak (valve2/3) test experiment



Comparison with baselines

Mean per-experiment ROC-AUC over all 34 experiments (threshold-free)

Method	Mean AUC	Median AUC	Experiments with AUC < 0.6
Time-RCD multivariate (used)	0.709	0.776	11 / 34
Time-RCD + 120 s moving average	0.710	0.798	12 / 34
Time-RCD univariate, mean of 8 channels	0.651	0.668	14 / 34
Isolation Forest (fit on anomaly-free)	0.555	0.509	27 / 34
Rolling z-score, 60 s window	0.545	0.545	32 / 34

- The pretrained model is far better than classical label-free baselines on the same data.
- Smoothing, ignoring the first 120 s of each file, and the univariate checkpoint did not help.
- Flow rate alone matched the multivariate model, but choosing a channel after seeing results would be test-set leakage, so it was not adopted.

Five representative test examples

Case	Experiment, time	Flow	Temp	Label	Score	Rank	Predicted
Normal, correctly normal	valve1/10, 13:34:39	32.0	67.7	0	0.004	0.146	normal
Anomaly, detected	other/8, 17:24:13	127.3	88.1	1	1.000	1.000	anomaly
Anomaly, missed	other/4, 17:20:44	76.0	87.6	1	0.012	0.001	normal
Normal, false alarm	other/14, 19:32:19	2.8	86.5	0	1.000	1.000	anomaly
Prolonged anomaly, detected	other/11, from 18:20:42	128.0	84.8	1	0.159	0.877	anomaly, 451 s in a row

- The miss (other/4, fluid leak): the sensors barely change at that second; the model's context sees nothing unusual.
- The false alarm (other/14): flow is 2.8, the pump is essentially off at the very start of the recording. The model treats the start-up transient as the most unusual event in the file.

Strengths, weaknesses, improvements

Strengths

- Zero-shot: no labels, no training, one call per file, CPU only.
- Handles all 8 sensors jointly.
- AUC 0.87 to 1.00 on most 'other' faults (rotor imbalance, leaks, water level).
- Clearly beats classical baselines.

Weaknesses

- Reliability varies: 11 of 34 files below 0.6 AUC, worst 0.21.
- Subtle valve faults (flow drop of about 1 unit) are missed; start-up transients fire instead.
- Score scale differs per recording, so rank normalisation is needed and one operational threshold is hard to set.
- Precision only 0.46 at the F1-optimal threshold.
- Domain gap: synthetic pretraining vs a small physical rig with step-like sensors.

Improvements

- Per-file threshold calibrated on a short known-normal warm-up window.
- Combine with a flow-rate rule for valve faults.
- Fine-tune on a few SKAB files with experiment-level cross-validation.
- Compare a second pretrained model (NVIDIA Kumo-Anomaly) on the same split.

Conclusion

- A pretrained zero-shot foundation model can be applied to industrial sensor data with a few lines of code and no training.
- On SKAB it detects most long faults and beats Isolation Forest and a rolling z-score by a wide margin (mean AUC 0.71 vs 0.55).
- It is not reliable enough on its own: a third of the experiments are near or below chance, subtle faults are missed, and precision is low.
- Honest answer to the research question: yes, it detects the faults, partly; a deployable system would need per-rig calibration or light fine-tuning.
- Everything is reproducible from the GitHub repository: one notebook, CPU only, about 15 minutes end to end.

Data: github.com/waico/SKAB (GPL-3.0) · Model: huggingface.co/thu-sail-lab/Time-RCD (Apache-2.0) · Code: github.com/asiralabi/skab-timercd-anomaly-detection